Psi_Sat Stage 4 — Wiring Guide (as built so far)

Board: Stage 4 EPS (EasyEDA Pro) · BQ25713 + TPS54302 · For Muhammad · Updated 2026-09-16
How to read this: Only things walked through and confirmed are marked done. Open / in-progress items are labeled. No guessing about unfinished nets.

1. What Stage 4 is for

Stage 4 is the power board in the CubeSat stack. It should:

- Take solar power in from the right header
- Charge / manage the battery pack that lives on Stage 5
- Make VSYS, then regulated 5 V and 3.3 V for the stack
- Carry I2C so the computer can talk to the charger
- Help fire the Stage 6 antenna burn wire (control + output path)

2. Big picture (simple flow)

Solar (right hdr) → 10 mΩ sense → VBUS → BQ25713 switcher (four FETs + 1 µH) → VSYS → battery path (10 mΩ + P-FET) / Stage 5 pack, and VSYS → two TPS54302 bucks → +5V0_SYS and +3V3_SYS.

3. Left 2×20 header (stack side)

Pin(s)	Net	What / why
1–2	GND	Common ground for the stack.
3–4	+3V3_SYS	3.3 V to the stack. Also feed the two 4.7 kΩ I2C pull-ups. Traces ~0.8 mm.
5–6	+5V0_SYS	5 V rail to the stack (from the 5 V TPS54302 / inductor pad 2).
7–8	VSYS	System rail from the charger / BATFET.
9–10	VBAT_CHG	Battery pack + from Stage 5 into Stage 4.
11–12	I2C SDA / SCL	Bus to BQ25713 (pins 12 / 13) with 4.7 kΩ pull-ups to 3.3 V.
13	BURN_CTRL	Burn command into AO3400A gate (via 10 kΩ). Logic only — not battery.
14–20+	NC	Unused on Stage 4 for now.

4. Right 2×20 header

Pin(s)	Net	What / why
1–2	GND	Ground.
3–4	V_SOLAR_POS	Solar input into the RAC 10 mΩ, then into charger VBUS.
7	BURN_OUT	Burn-wire side from the burn N-FET drain toward Stage 6.
Other	NC	Nothing else used on the right header so far.

Note: early walkthrough also mentioned burn on right pin 13; you later said pin 7. Match silkscreen to copper.

5. Battery charge path

Item	Connection	Why
10 mΩ pad 2	Hdr 9/10 + BQ25713 pin 19 (SRN)	Pack side of sense resistor.

10 mΩ pad 1	BQ25713 pin 20 (SRP) + BATFET drain net	Charger side of sense resistor.
AO3401A Gate	BQ25713 pin 21 (BATDRV)	Chip opens/closes battery↔system path.
AO3401A Source	VSYS (pin 22)	System side of BATFET.
AO3401A Drain	10 mΩ pad 1 net	Battery/sense side of BATFET.

6. Solar input path

Item	Connection	Why
Hdr 3–4	RAC 10 mΩ pad B (solar side)	Solar power enters here.
RAC pad A	VBUS pin 1 + ACN pin 2 + ~10 µF	Main current into the charger input.
ACP pin 3	RAC pad B only (sense)	Measures input current — does not carry amps.
ACN pin 2	RAC pad A / VBUS (sense)	Other side of that measurement.

Removed wrong wire that dumped solar into pin 31 (HIDRV1). Pin 31 is gate-drive only.

7. Burn path

Item	Connection	Why
AO3400A Gate (1)	BURN_CTRL (left hdr 13) via 10 kΩ	MCU/stack turns burn FET on/off.
AO3400A Source (2)	GND	Low-side switch return.
AO3400A Drain (3)	BURN_OUT (right hdr 7)	Goes toward burn wire / Stage 6.

8. Charger setup parts (bias / config)

Item	Wiring	Why
CELL_BAT pin 18	REGN pin28 —150 kΩ— pin18 —100 kΩ— GND	Tells chip: 2S battery.
Parts	150 kΩ C22807; 100 kΩ C25803	LCSC parts used.
VDDA pin 7	REGN —10 Ω— pin7 —1 µF— GND	Internal clean bias supply.
REGN pin 28 cap	pin28 —10 µF (C15850)— GND	Required REGN bulk cap.
PGND pin 27	GND (fat)	Power ground for the switcher.
COMP1 / COMP2	TI-style RC on pins 16 / 17	Loop compensation.

Later tweak: with 1 µH inductor, COMP R23 ≈ 4.7 kΩ (optional). Bootstrap caps still open (see §10).

9. Buck-boost power stage (FETs + inductor)

Inductor (confirmed): pin 32 (SW1) — 1 µH (CYA0630-1.0UH) — pin 23 (SW2).

FET	Role	Wiring (AO3400A C347475)	Status
Q1	Buck high-side	D→VBUS pin1; S→SW1 pin32; G→HIDRV1 pin31	Done
Q2	Buck low-side	D→SW1 pin32; S→GND; G→LODRV1 pin29	Done
Q3	Boost low-side	D→SW2 pin23; S→GND; G→LODRV2 pin26	Done
Q4	Boost high-side	D→VSYS pin22; S→SW2 pin23; G→HIDRV2 pin24	Done

AO3400A: 1=Gate, 2=Source, 3=Drain.

10. Still open on charger FETs

- Bootstrap caps: BTST1 (pin 30)↔SW1, BTST2 (pin 25)↔SW2 (~0.047 µF each)
- Optional: COMP R23 ≈ 4.7 kΩ for 1 µH

11. 5 V buck (TPS54302) — as you described

IC: TPS54302DDCR (C311983). Inductor: **same part both bucks** — SRG0630-4R7-M / **C53462987** (4.7 µH).

Item	Connection	Why / notes
Inductor pad 1	TPS pin 2 (SW) = NET_SW_5V	Switch node into inductor.
100 nF	TPS pin 6 (BOOT) ↔ pin 2 (SW)	Bootstrap capacitor.
Inductor pad 2	+5V0_SYS	5 V output node.
From +5V0_SYS	Left hdr pins 5–6	Feeds the stack.
From +5V0_SYS	100 kΩ → TPS pin 4 (FB)	Top of feedback divider (NET_FB_5V).
FB → GND	150 Ω (as read on board)	Bottom of divider — recheck value (see note).
TPS pins 3 & 5	Tied together on VSYS	VIN + EN always-on from VSYS.
VIN bypass	1 µF on pin 3 (VIN)	Input cap (add more bulk later if needed).

FB note: with TPS54302 Vref ≈ 0.596 V, 100 kΩ / 150 Ω would not target 5 V. Double-check the bottom resistor value on the board before spin.

12. 3.3 V buck (TPS54302) — in progress

Same IC family and same inductor C53462987. Pad 2 net is named +3V3_SYS. You reported pad 2 currently only goes to one resistor pad — mirror of the 5 V twin is incomplete.

Item	Connection	Status
Inductor (C53462987)	Same 4.7 µH as 5 V buck	On board
Pad 2 net name	+3V3_SYS	Named
Pad 2 → resistor	One resistor pad so far	Partial
Pad 2 → left hdr 3–4	Should feed stack 3.3 V	Step 1 — do this now
BOOT–SW 100 nF	Like 5 V buck	Not walked yet
FB divider complete	Top from +3V3_SYS, bottom to GND	Not walked yet
VIN/EN from VSYS + caps	Mirror 5 V buck	Not walked yet

Current step (Sep 2026 session): from inductor pad 2 / +3V3_SYS, also wire left header pins 3–4 (keep the existing resistor wire).

Update covers Q1–Q4 done, 5 V buck as described by you, and 3.3 V buck work in progress. Refresh again after 3.3 V steps and bootstrap caps finish.